Santiago Montaña

Data Scientist | Data & AI Engineer | Conversational AI & Cloud Operations

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PROFESSIONAL SUMMARY

Data Scientist and Data & AI Engineer with a degree in Electronic Engineering (2024) and a Graduate Specialization in Strategic Data Analytics (UPTC, 2025). Proven track record designing and orchestrating end-to-end ETL/ELT data pipelines (Snowflake, Apache Airflow, dbt, Meltano), training predictive models and ensemble classifiers (Random Forest, XGBoost, Monte Carlo quantitative simulations tracked in MLflow), building production conversational AI & autonomous agents (LLMs, Model Context Protocol - MCP, ElevenLabs voice AI), and provisioning reproducible cloud infrastructure on AWS via Terraform (IaC), Docker, and FastAPI under CI/CD workflows.

PROFESSIONAL EXPERIENCE

Data Scientist & AI Developer

Jan 2026 – Present

CIM DATA IA — Consultoría de Datos e Inteligencia Artificial — Bogotá / Remoto, Colombia

- Trained and deployed predictive models and ensemble classifiers in Python (scikit-learn, Random Forest, XGBoost) tracked with MLflow, improving operational forecasting and prioritization accuracy by +20%.
- Engineered conversational AI solutions and autonomous agents integrating Model Context Protocol (MCP), automating enterprise workflows and reducing manual operational intervention by ~35%.
- Provisioned and standardized cloud infrastructure on AWS (EC2, S3, IAM) using Terraform (IaC), achieving 100% environment reproducibility and cutting deployment lead times by ~45%.
- Executed quantitative Monte Carlo simulations (+10,000 scenarios) and stress tests to validate data architecture resilience, orchestrating automated ETL/ELT pipelines with Python, SQL, Snowflake, dbt, and Apache Airflow under CI/CD.

Data & AI Development Associate

May 2025 – Dec 2025

CIM DATA IA — Soluciones Financieras y Conversacionales — Bogotá / Boyacá, Colombia

- Analytically tuned and optimized banking conversational AI models and NLP intent pipelines, increasing intent classification accuracy by +22% and reducing fallback rates in client interactions.
- Co-developed a low-latency AI voice assistant for automated loan verification in the financial sector using Python, ElevenLabs, and third-party REST APIs, cutting verification cycle time by ~40%.
- Automated recurring operational data workflows and standardized reproducible Python environments across the engineering team, reducing environment setup errors by ~50%.

Data & Industrial Operations Automation Intern

May 2024 – Nov 2024

HOLCIM S.A. — Operaciones Industriales y Planta — Nobsa, Boyacá, Colombia

- Automated industrial plant data ingestion, cleaning, and validation using Python scripts and Google Apps Script, reducing manual reporting effort by ~40%.
- Designed and deployed a centralized relational database to track +500 industrial assets and maintenance records, eliminating data silos across plant operations.
- Built interactive executive dashboards in Looker Studio to monitor real-time plant productivity, equipment availability (OEE), and operational KPIs for plant leadership.

TECHNICAL SKILLS

Data Engineering & Orchestration:	Snowflake (Cloud Data Warehouse), Apache Airflow (DAG Orchestration), dbt (SQL Transformations & Data Modeling), Meltano (ELT Extraction), Pipelines ETL / ELT & Data Quality
Machine Learning & Data Science:	Scikit-learn (Random Forest, XGBoost, Ensembles), Simulaciones Cuantitativas (Monte Carlo), MLflow (Experiment Tracking & MLOps), Análisis Estadístico & Validación de Hipótesis, Feature Engineering, Pronóstico & Clasificación
Conversational AI & MCP Agents:	Model Context Protocol (MCP) Integration, Agentes Autónomos de IA & Workflows, LLMs (Claude, OpenAI, Prompt Engineering), ElevenLabs (Asistentes de Voz de Baja Latencia), Optimización de Chatbots Bancarios & NLP
Cloud, DevOps & IaC:	Terraform (Infrastructure as Code - IaC), AWS (EC2, S3, IAM), FastAPI & Pydantic (APIs REST en Producción), Docker & Entornos Reproducibles, CI/CD con GitHub Actions & Git

Programming Languages:	Python (pandas, NumPy, SciPy, scikit-learn, Matplotlib), SQL (Consultas Analíticas, CTEs, Window Functions, Optimización), JavaScript / TypeScript (ES6+)
Databases & Warehousing:	Snowflake, PostgreSQL, MySQL, SQLite
Visualization & BI:	Looker Studio, Power BI, Tableau, Storytelling con Datos & KPIs Operativos

FEATURED PROJECTS

Supply Chain & Marketplace Analytics (Coomproriente) *Stack: Python, pandas, SQL, Data Visualization, Operations Analytics*

Quantitative modeling of carriers, merchants, and pricing dynamics to optimize regional supply chain logistics and marketplace operations.

Autonomous News & Media AI Agent *Stack: Python, LLMs, MCP, Autonomous Agents, Automation*

Autonomous AI pipeline integrating LLMs and MCP workflows for automated research, synthesis, and multi-channel content generation.

Multiclass Vehicle Classification API & Web App *Stack: Python, scikit-learn, FastAPI, Pydantic, Streamlit*

End-to-end classification model serialized and deployed in production with FastAPI, Pydantic schema validation, structured logging, automated tests, and Streamlit UI.

USD/COP Exchange Rate (TRM) & Crypto Monte Carlo Predictor *Stack: Python, Scikit-learn, Monte Carlo, NLP Sentiment, Time Series*

Predictive modeling combining historical time series, real-time news NLP sentiment analysis, and Monte Carlo stochastic simulations (+10,000 paths) to estimate market scenarios.

EDUCATION

Specialization in Strategic Data Analytics
Universidad Pedagógica y Tecnológica de Colombia (UPTC) · Graduate Specialization · Graduated 2025

Bachelor of Science in Electronic Engineering
Universidad Pedagógica y Tecnológica de Colombia (UPTC) · B.Sc. Degree · Graduated 2024

LANGUAGES

Spanish — Native (C2)
English — Upper-Intermediate (B2 Professional)

CERTIFICATIONS & COURSES

MLOps & Despliegue de Modelos de Machine Learning	2025	Data Scientist con Python	Apr 2025
Decision Trees y Random Forest con scikit-learn	Mar 2025	Tableau — Visualización de Datos y Storytelling	Mar 2025
Power BI para Analítica de Negocios	2025	Regresión Lineal con Python y scikit-learn	Feb 2025
Arquitectura y Proyectos de Ciencia de Datos	Feb 2025	Git, GitHub Profesional & Flujos CI/CD	Jan 2025
Bases de Datos Relacionales y Consultas con SQL	Sep 2024	Python Aplicado a Ciencia de Datos	Sep 2024